

NoSQL Unit 4

UNIT IV — Topic Wise Question Bank

Cassandra: Architecture & Replication Factor Strategy

1 Mark Questions

Q1.

Which of the following is NOT a common data model used in NoSQL databases?

— 1 Mark

- a) Key-value store
 - b) Column-family store
 - c) Document store
 - d) Relational table store
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Cassandra: Features

No direct questions found from the uploaded papers.

Cassandra Keyspace — Create, Alter & Drop Keyspace in Cassandra

No direct questions found from the uploaded papers.

Cassandra Table — Create, Alter, Drop & Truncate

No direct questions found from the uploaded papers.

Redis: Introduction

4 Mark Questions

Q1.

What is Redis? Describe its main data types — String, List, Set, Hash, and Sorted Set — and provide an example for each. — 4 Marks

Redis: Features

6 Mark Questions

Q1.

Describe the various Redis data types. For each type, mention a use case where it would be most suitable. — 6 Marks

Redis: Datatypes

1 Mark Questions

Q1.

Which REDIS datatype is best suited to store a collection of unique values? — 1 Mark

- a) string
- b) list

- c) set
 - d) hash
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Redis: Keys

No direct questions found from the uploaded papers.

Redis: String

No direct questions found from the uploaded papers.

Graph Databases & Neo4j

1 Mark Questions

Q1.

What is a key feature of graph databases? — 1 Mark

- a) fixed schema
 - b) efficient storage and querying of relationships
 - c) store data only in tabular format
 - d) use of SQL for all queries
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Q2.

Which neo4j clause is used to remove properties or labels from nodes? — 1 Mark

- a) delete
 - b) remove
 - c) foreach
 - d) limit
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Q3.

Which is the best definition for the graph database? — 1 Mark

- A) A system that organizes data in collections and documents.
 - B) A system that stores information in tabular formats.
 - C) A database built on the concept of graph theory using nodes and edges.
 - D) A relational database system optimized for joins.
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Q4.

What makes Neo4j distinct as a database system? — 1 Mark

- A) It uses JSON-based querying.
 - B) It integrates natively with MySQL.
 - C) It leverages graph structures and the Cypher query language.
 - D) It's primarily used for time-series analysis.
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Q5.

In Cypher (Neo4j), which clause is used to find patterns in the graph? — 1 Mark

- A) MATCH
 - B) FIND
 - C) QUERY
 - D) SEARCH
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Q6.

Why would a Neo4j node have both :Customer and :VIP labels? — 1 Mark

- A) To increase indexing performance by duplicating labels
 - B) To allow querying the node as either :Customer or :VIP
 - C) Because each node must have a minimum of two labels
 - D) To prevent the node from being matched in generic queries
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HiveQL (GROUP BY / HAVING / JOIN)

1 Mark Questions

Q1.

In HiveQL, which clause is used to filter groups after aggregation? — 1 Mark

- a) where
 - b) having
 - c) group by
 - d) order by
-

Q2.

In HiveQL, what is the main difference between JOIN and GROUP BY? — 1 Mark

- a) JOIN combines rows from two or more tables; GROUP BY aggregates rows based on column values
 - b) JOIN filters rows; GROUP BY sorts rows
 - c) JOIN sorts rows; GROUP BY combines rows
 - d) JOIN aggregates rows; GROUP BY filters rows
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HBase

1 Mark Questions

Q1.

In HBase, what is the main use of the SCAN operation? — 1 Mark

- A) To write a new row to the table.
- B) To fetch a single row using its row key.
- C) To iterate over rows in a table based on filter criteria.
- D) To delete rows from a table.

Q2.

Which filter in HBase is suitable for selecting rows where a specific column's value exceeds 100? — 1 Mark

- A) FamilyFilter
 - B) SingleColumnValueFilter
 - C) ColumnPaginationFilter
 - D) RowFilter
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Hive

1 Mark Questions

Q1.

What is the correct HiveQL command to define a new table schema in Hive? — 1 Mark

- A) CREATE SCHEMA TABLE
 - B) DEFINE TABLE
 - C) CREATE TABLE
 - D) MAKE DATABASE
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Q2.

Choose the correct statement about Apache Hive: — 1 Mark

- A) Hive is optimized for transactional workloads.
 - B) Hive allows SQL-like queries on large-scale data stored in Hadoop.
 - C) Hive runs independently of the Hadoop ecosystem.
 - D) Hive stores data in key-value formats like NoSQL systems.
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Q3.

What is the result of executing the Hive query: `SELECT COUNT(*) FROM departments;`? — 1 Mark

- A) A list of departments with employee counts
 - B) The total number of department names
 - C) A detailed record of all departments
 - D) The number of rows in the departments table
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Q4.

How would you calculate the average age of employees in each department using Hive? — 1 Mark

- A) `SELECT AVG(age), department FROM employees;`
 - B) `SELECT department FROM employees WHERE AVG(age);`
 - C) `SELECT department, AVG(age) FROM employees GROUP BY department;`
 - D) `SELECT department, COUNT(age) FROM employees GROUP BY department;`
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6 Mark Questions

Q1.

Explain how `GROUP BY` and `HAVING` clauses are used in Hive. Write a query to get the departments having more than 5 students. — 6 Marks

Graph Databases — Long Answer Questions

4 Mark Questions

Q1.

Explain the fundamental idea behind graph databases and highlight the key differences between graph databases and relational database systems. — 4 Marks

Q2.

Explain how you can store a simple project management graph (tasks and their dependencies) in a graph database. Write a query to find all tasks that directly depend on a given task. — 4 Marks

Q3.

Write a query to create and find all Product nodes with a price greater than 100 and category as "Electronics." — 4 Marks

Q4.

Write a query to create and find all Employee nodes with a salary greater than 70,000 who have at least one: MANAGES relationship to a Project node name "Apollo." — 4 Marks

6 Mark Questions

Q1.

Compare Relational Databases, Graph Databases, Key-Value Stores, and Column-Family Stores focusing on their schema design, query languages, use cases and performance. — 6 Marks

Q2.

Explain the fundamental idea behind graph databases and highlight the key differences between graph databases and relational database systems. — 6 Marks

Q3.

Write a Cypher query using Neo4j to create a node using the SET, SKIP, Aggregation clause. Explain the syntax and effect of the query. — 6 Marks

Q4.

Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order. — 6 Marks